



Institute of Technology of Cambodia

Dept. of Telecommunication and Network Engineering

4th Year, 7th Promotion

“Routing and Switching ”

Enterprise Network

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Academic Year: 2026-2027

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I. INTRODUCTION AND EXPERIMENT OBJECTIVES

1.1. Introduction

This project focuses on the design and implementation of an enterprise network using **Cisco Packet Tracer**. The network is divided into two locations:

- **Head Office**
- **Branch Office**

Each location contains several departments connected through routers and switches. To improve network performance and security, every department is assigned to a separate VLAN.

The Head Office consists of:

- HR Department
- IT Department
- Server Department

The Branch Office consists of:

- GTR-B
- GTR-A
- Library
- Staff
- CCTV

Besides end-user devices, the network also includes:

- Routers
- Layer 2 Switches
- Wireless Access Points
- PCs and Laptops
- Web Server
- File Server

To allow communication between the Head Office and Branch Office, **OSPF dynamic routing** is configured on the routers. OSPF automatically exchanges routing information, reducing manual configuration and improving scalability.

On the switching side, **Spanning Tree Protocol (STP)** is implemented in the Head Office to prevent Layer 2 loops and ensure network stability when redundant links are present.

For network security, **Access Control Lists (ACLs)** are applied to control communication between VLANs. ACL rules determine which departments are allowed or denied access to

specific servers and networks. This helps protect sensitive resources such as the File Server while still allowing authorized users to access the Web Server and other permitted services.

Overall, this project demonstrates the integration of routing, switching, VLAN segmentation, dynamic routing, network security, and wireless connectivity in a simulated enterprise environment.

1.2. Project Objectives

The main objective of this project is to design and implement a secure, reliable, and scalable enterprise network using Cisco Packet Tracer. The project applies routing and switching technologies to enable communication between departments while maintaining network performance and security.

The objectives of this project are:

- Design an enterprise network topology for the Head Office and Branch Office.
- Create VLANs to separate departments into different broadcast domains.
- Configure IP addressing and default gateways for all VLANs.
- Configure trunk links to allow multiple VLANs to communicate between switches and routers.
- Configure Inter-VLAN Routing to enable communication between different VLANs.
- Implement OSPF dynamic routing to exchange routing information automatically between routers.
- Configure Spanning Tree Protocol (STP) to prevent switching loops and improve network reliability.
- Configure Web Server and File Server services for network users.
- Configure wireless access points to provide wireless connectivity for laptop users.
- Apply Access Control Lists (ACLs) to enforce network security policies.
- Test and verify network connectivity to ensure all configurations work correctly.

1.3. Project Scope

The scope of this project covers the design, implementation, and testing of an enterprise network using Cisco Packet Tracer. The project focuses on the routing and switching technologies learned throughout the course.

The project includes the following components:

Head Office

- HR VLAN
- IT VLAN

- Server VLAN
- Spanning Tree Protocol
- Web Server
- File Server

Branch Office

- GTR-B VLAN
- GTR-A VLAN
- Library VLAN
- Staff VLAN
- CCTV VLAN
- Wireless Access Points

Network Configuration

The project covers the implementation of:

- VLAN Configuration
- Access Port Configuration
- Trunk Configuration
- Inter-VLAN Routing
- OSPF Dynamic Routing
- Spanning Tree Protocol (STP)
- Server Configuration
- Wireless Configuration
- Access Control Lists (ACL)

Testing

The project also includes testing and verification, including:

- VLAN verification
- Trunk verification
- OSPF neighbor verification
- Routing table verification
- STP verification
- ACL verification
- Ping connectivity tests
- Web Server access tests
- File Server access tests

II. Methodology

This project was implemented using Cisco Packet Tracer. The configuration was completed step by step, starting from the physical topology design, VLAN creation, IP addressing, trunk configuration, inter-VLAN routing, OSPF dynamic routing, Spanning Tree Protocol, server configuration, ACL security rules, and final connectivity testing.

The main devices used in this project are routers, Cisco 2960 switches, PCs, laptops, wireless access points, a web server, and a file server. The network is divided into a Head Office and a Branch Office. The Head Office contains HR, IT, and Server VLANs, while the Branch Office contains GTR-B, GTR-A, Library, Staff, and CCTV VLANs.

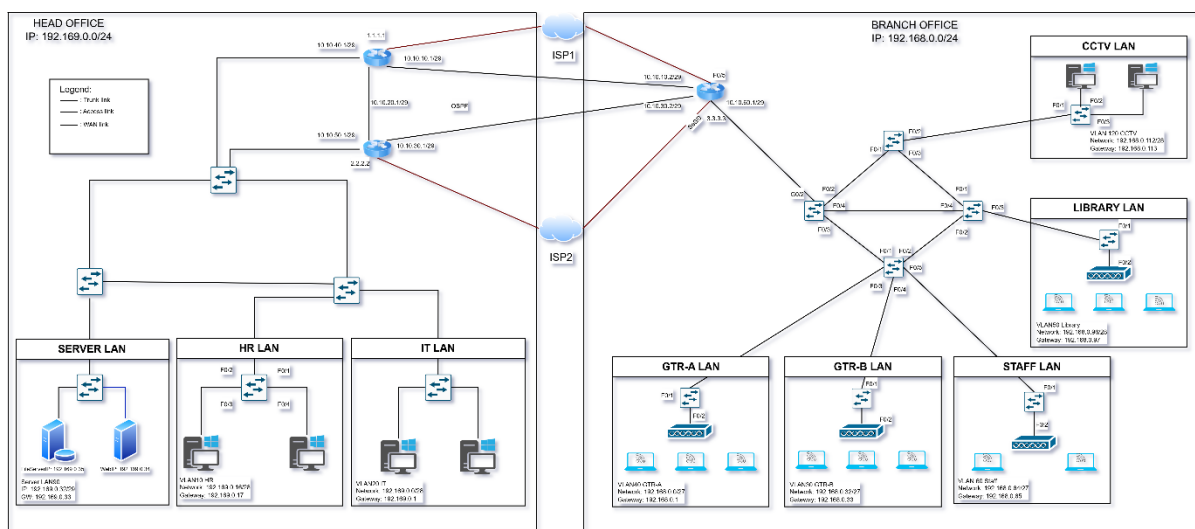


Figure II.1 Enterprise Network Topology

III. IP Addressing and VLAN Design

Each department was assigned to a separate VLAN and subnet. VLANs help separate departments logically, reduce broadcast traffic, and improve security.

Table III.1 VLAN and IP Addressing Design

VLAN	Department	Network	Gateway
VLAN 10	HR	192.169.0.16/28	192.169.0.17
VLAN 20	IT	192.169.0.0/28	192.169.0.1
VLAN 90	Server	192.169.0.32/29	192.169.0.33
VLAN 30	GTR-B	192.168.0.0/27	192.168.0.1
VLAN 40	GTR-A	192.168.0.32/27	192.168.0.33
VLAN 50	Library	192.168.0.96/28	192.168.0.97
VLAN 60	Staff	192.168.0.64/27	192.168.0.65

VLAN 120	CCTV	192.168.0.112/28	192.168.0.113
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IV. VLAN Configuration

VLAN, or Virtual Local Area Network, is a technology used to divide one physical network into several logical networks. Devices in different VLANs are separated even if they are connected to the same switch. VLAN helps improve security, reduce broadcast traffic, and organize users by department.

4.1. Creating VLANs on Head Office Switch

In the Head Office, VLAN 10, VLAN 20, and VLAN 90 were created for HR, IT, and Server networks.

Configuration:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# name GTR-B
Switch(config-vlan)# exit
Switch(config)# vlan 40
Switch(config-vlan)# name GTR-A
Switch(config-vlan)# exit
Switch(config)# vlan 50
Switch(config-vlan)# name LIBRARY
Switch(config-vlan)# exit
Switch(config)# vlan 60
Switch(config-vlan)# name STAFF
Switch(config-vlan)# exit
Switch(config)# vlan 120
Switch(config-vlan)# name CCTV
Switch(config-vlan)# exit
```

```
S1-Head#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1
10	VLAN0010	active	
20	VLAN0020	active	
90	VLAN0090	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1-Head# |
```

Figure IV.1 Head Office VLAN Creation Verification

4.2. Creating VLANs on Branch Office Switch

In the Branch Office, VLAN 30, VLAN 40, VLAN 50, VLAN 60, and VLAN 120 were created for GTR-B, GTR-A, Library, Staff, and CCTV networks.

Configuration command:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# name GTR-B
Switch(config-vlan)# exit
Switch(config)# vlan 40
Switch(config-vlan)# name GTR-A
Switch(config-vlan)# exit
Switch(config)# vlan 50
Switch(config-vlan)# name LIBRARY
Switch(config-vlan)# exit
Switch(config)# vlan 60
Switch(config-vlan)# name STAFF
Switch(config-vlan)# exit
Switch(config)# vlan 120
Switch(config-vlan)# name CCTV
Switch(config-vlan)# exit
Switch# show vlan brief
```

```
S1-Branch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1
30	VLAN0030	active	
40	VLAN0040	active	
50	VLAN0050	active	
60	VLAN0060	active	
120	VLAN0120	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1-Branch#
```

Figure IV.2 Branch Office VLAN Creation Verification

4.3. Assigning Access Ports

After creating VLANs, access ports were configured for PCs, servers, laptops, and access points. Each access port belongs to only one VLAN.

Configuration command:

```
Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Figure IV.3 Access Port Assignment Verification

For IT VLAN and Server VLAN use command show vlan brief to verify access port in switch of each vlan.

4.4. Trunk Configuration

Trunk ports were configured between switches and between the switch and router. A trunk link allows traffic from multiple VLANs to pass through one physical connection using 802.1Q tagging.

Configuration command:

```
S1-Head#show interfaces trunk
Port          Mode          Encapsulation  Status        Native vlan
Fa0/3         on            802.1q         trunking      1
Fa0/4         on            802.1q         trunking      1
Gig0/2        on            802.1q         trunking      1

Port          Vlans allowed on trunk
Fa0/3         10,20,90
Fa0/4         10,20,90
Gig0/2        10,20,90

Port          Vlans allowed and active in management domain
Fa0/3         10,20,90
Fa0/4         10,20,90
Gig0/2        10,20,90

Port          Vlans in spanning tree forwarding state and not pruned
Fa0/3         10,20,90
Fa0/4         10,20,90
Gig0/2        10,20,90
```

Figure IV.4 Trunk Port Verification

V. Inter-VLAN Routing Configuration

A router sub-interface(inter vlan) is a virtual interface created on one physical router interface. Each sub-interface is assigned to one VLAN and uses 802.1Q encapsulation. This method is commonly used in Router-on-a-Stick configuration.

Configuration command in head office:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.169.0.17 255.255.255.240
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.169.0.1 255.255.255.240
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.90
Router(config-subif)# encapsulation dot1Q 90
Router(config-subif)# ip address 192.169.0.33 255.255.255.248
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Configuration command in branch office:

```
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.0.1 255.255.255.224
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.0.33 255.255.255.224
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.50
Router(config-subif)# encapsulation dot1Q 50
Router(config-subif)# ip address 192.168.0.97 255.255.255.240
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.60
Router(config-subif)# encapsulation dot1Q 60
Router(config-subif)# ip address 192.168.0.65 255.255.255.224
Router(config-subif)# exit
Router(config)# interface gigabitEthernet 0/0.120
Router(config-subif)# encapsulation dot1Q 120
Router(config-subif)# ip address 192.168.0.113 255.255.255.240
Router(config-subif)# exit
```

```
Router3#show ip interface brief
Interface          IP-Address      OK? Method Status Protocol
FastEthernet0/0    10.10.10.2      YES manual up      up
FastEthernet1/0    unassigned      YES unset  down    down
FastEthernet2/0    10.10.30.2      YES manual up      up
FastEthernet3/0    10.10.60.1      YES manual up      up
FastEthernet3/0.30 192.168.0.1     YES manual up      up
FastEthernet3/0.40 192.168.0.33    YES manual up      up
FastEthernet3/0.50 192.168.0.97    YES manual up      up
FastEthernet3/0.60 192.168.0.65    YES manual up      up
FastEthernet3/0.120 192.168.0.113   YES manual up      up
Serial4/0          unassigned      YES unset  administratively down down
Router3#
```

Figure V.1 Router Sub-Interface in branch verification

```
Router1#show ip interface brief
Interface          IP-Address      OK? Method Status Protocol
FastEthernet0/0    10.10.10.1      YES manual up      up
FastEthernet1/0    10.10.20.1      YES manual up      up
FastEthernet2/0    10.10.40.1      YES manual up      up
FastEthernet2/0.10 192.169.0.17    YES manual up      up
FastEthernet2/0.20 192.169.0.1     YES manual up      up
FastEthernet2/0.90 192.169.0.33    YES manual up      up
FastEthernet3/0    unassigned      YES unset  down    down
Serial4/0          unassigned      YES unset  administratively down down
Router1#
```

Figure V.2 Router Interface head office Verification

VI. OSPF Dynamic Routing Configuration

OSPF, or Open Shortest Path First, is a dynamic routing protocol used by routers to exchange routing information automatically. It calculates the best path based on cost and updates the routing table when the network changes.

Configuration command for router head office:

```
Router1(config)# router ospf 1
Router1(config-router)# router-id 1.1.1.1
Router1(config-router)# network 10.10.10.0 0.0.0.7 area 0
Router1(config-router)# network 10.10.20.0 0.0.0.7 area 0
Router1(config-router)# network 10.10.40.0 0.0.0.7 area 0
Router1(config-router)# network 192.169.0.0 0.0.0.63 area 0
Router1(config-router)# exit
```

```
Router1(config)# router ospf 1
Router2(config-router)# router-id 2.2.2.2
Router2(config-router)# network 10.10.20.0 0.0.0.7 area 0
Router2(config-router)# network 10.10.30.0 0.0.0.7 area 0
Router2(config-router)# network 10.10.50.0 0.0.0.7 area 0
Router2(config-router)# network 192.169.0.0 0.0.0.63 area 0
Router2(config-router)# exit
```

Configuration command for router branch office:

```
Router1(config)# router ospf 1
Router2(config-router)# router-id 3.3.3.3
Router2(config-router)# network 10.10.10.0 0.0.0.7 area 0
Router2(config-router)# network 10.10.30.0 0.0.0.7 area 0
Router2(config-router)# network 10.10.60.0 0.0.0.7 area 0
Router2(config-router)# network 192.168.0.0 0.0.0.127 area 0
Router2(config-router)# exit
```

Verify OSPF neighbor relationship in router 1:

```
Router1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/DR         00:00:31    10.10.10.2     FastEthernet0/0
2.2.2.2        1     FULL/DR         00:00:31    10.10.20.2     FastEthernet1/0
Router1#
```

Figure VI.1 OSPF Neighbor Verification Router 1

```
Router2#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/DR         00:00:33    10.10.30.2     FastEthernet2/0
1.1.1.1        1     FULL/BDR        00:00:34    10.10.20.1     FastEthernet1/0
Router2#
```

Figure VI.2 OSPF Neighbor Verification Router 2

```
Router3#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
1.1.1.1        1     FULL/BDR        00:00:36    10.10.10.1     FastEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:36    10.10.30.1     FastEthernet2/0
Router3#
```

Figure VI.3 OSPF Neighbor Verification Router 3

VII. Spanning Tree Protocol Configuration

Spanning Tree Protocol, or STP, is a Layer 2 switching protocol used to prevent network loops. When there are redundant switch links, STP blocks one path and keeps another path active. If the active path fails, the backup path can be used.

Configuration command:

```
Switch> enable
Switch# configure terminal
Switch(config)# spanning-tree mode rapid-pvst
Switch(config)# exit
```

To make the main Head Office switch become the root bridge:

```
Switch(config)# spanning-tree vlan 10,20,90 root primary
```

Verify STP:

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0002.4AA1.47A4
             Cost        38
             Port        2 (FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0060.5C57.EC57
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Altn BLK 19       128.1    P2p
Fa0/2                    Root FWD 19       128.2    P2p
```

Figure VII.1 Spanning Tree VLAN0001 Configuration

```
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0005.5E70.7E0E
             Cost        19
             Port        1 (FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0060.5C57.EC57
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19       128.1    P2p
Fa0/2                    Altn BLK 19       128.2    P2p
```

Figure VII.2 Spanning Tree VLAN0010 Configuration

VLAN0020					
Spanning tree enabled protocol ieee					
Root ID	Priority	24596			
	Address	0005.5E70.7E0E			
	Cost	19			
	Port	1(FastEthernet0/1)			
	Hello Time	2 sec	Max Age	20 sec	Forward Delay 15 sec
Bridge ID	Priority	28692 (priority 28672 sys-id-ext 20)			
	Address	0060.5C57.EC57			
	Hello Time	2 sec	Max Age	20 sec	Forward Delay 15 sec
	Aging Time	20			
Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p

Figure VII.3 Spanning Tree VLAN0020 Configuration

VLAN0090					
Spanning tree enabled protocol ieee					
Root ID	Priority	32769			
	Address	0060.4768.13B5			
	Cost	19			
	Port	3(FastEthernet0/3)			
	Hello Time	2 sec	Max Age	20 sec	Forward Delay 15 sec
Bridge ID	Priority	32858 (priority 32768 sys-id-ext 90)			
	Address	0060.5C57.EC57			
	Hello Time	2 sec	Max Age	20 sec	Forward Delay 15 sec
	Aging Time	20			
Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/3	Root	FWD	19	128.3	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

Figure VII.4 Spanning Tree VLAN0090 Configuration

VIII. Server VLAN

A Server VLAN is a separate VLAN used to place important servers in one protected network segment. This makes it easier to manage and secure server access.

Web Server configuration:

IP Address: 192.169.0.34
Subnet Mask: 255.255.255.248
Default Gateway: 192.169.0.33

File Server configuration:

IP Address: 192.169.0.35
Subnet Mask: 255.255.255.248
Default Gateway: 192.169.0.33

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:70FF:FE50:3E00
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.169.0.35
    Subnet Mask . . . . .: 255.255.255.248
    Default Gateway . . . . .: ::
                                192.169.0.33

```

Figure VIII.1 Web Server IP Configuration

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FE81:E75A
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.169.0.34
    Subnet Mask . . . . .: 255.255.255.248
    Default Gateway . . . . .: ::
                                192.169.0.33

```

Figure VIII.2 File Server IP Configuration

HTTP

HTTP
☒ On ☐ Off

HTTPS
☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

Figure VIII.3 Web Server HTTP Service

IX. Wireless Access Point Configuration

Wireless access points were configured for laptop users in GTR-B and GTR-A. The access points were connected to the correct switch ports and assigned to the correct VLANs. The access point connected to the GTR-B network belongs to VLAN 30. The access point connected to the GTR-A network belongs to VLAN 40.

Switch port configuration for GTR-B AP:

```
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
```

Switch port configuration for GTR-A AP:

```
Switch(config)# interface fastEthernet 0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# exit
```

On each laptop, the wireless network was selected and the correct IP address, subnet mask, and default gateway were configured. Wireless access allows laptop users to communicate with the enterprise network without using physical cables.

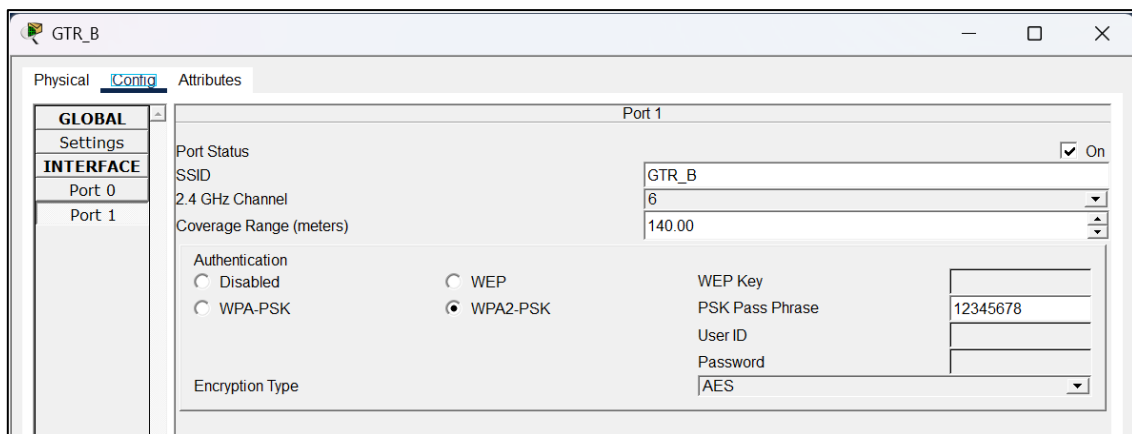


Figure IX.1 Access Point Connection

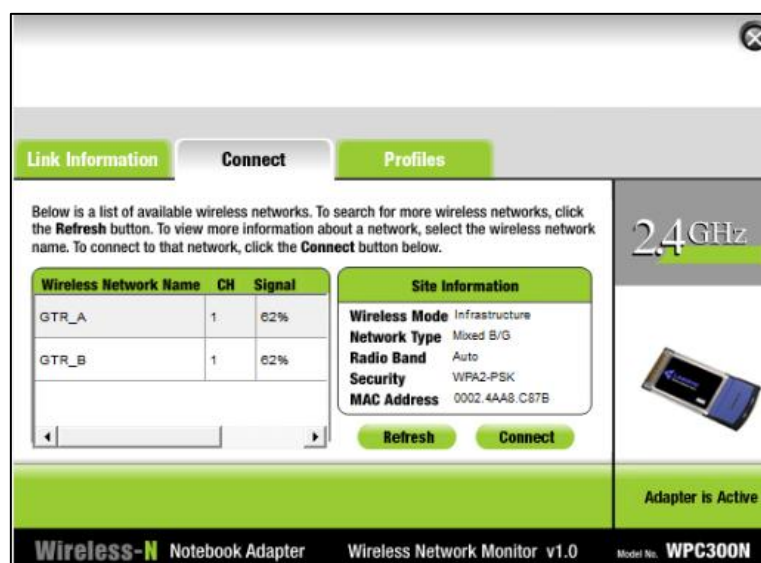


Figure IX.2 Laptop Wireless Connection

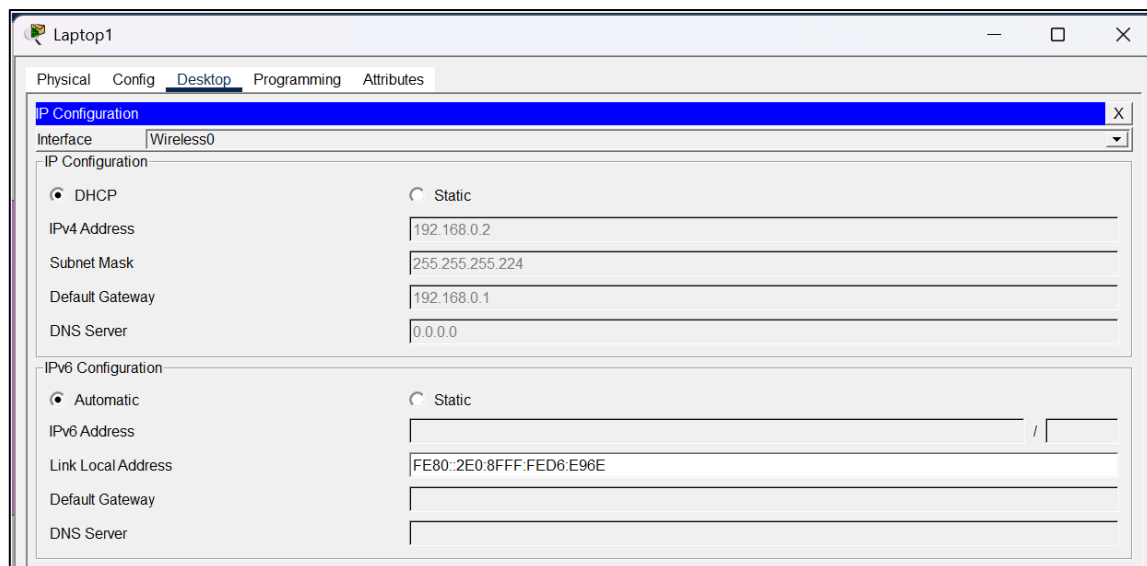


Figure IX.3 Wireless Client IP Configuration

X. Access Control List Configuration

ACL, or Access Control List, is a security feature used to filter network traffic. It can permit or deny traffic based on source IP address, destination IP address, protocol, or port number.

10.1. Head Office ACL

The Head Office ACL controls HR and IT access. HR is blocked from accessing the IT VLAN and the File Server. However, HR can access the Web Server and other permitted networks. IT is allowed to access all VLANs and servers.

ACL for HR:

```
Router(config)# ip access-list extended ACL_HR
Router(config-ext-nacl)# deny ip 192.169.0.16 0.0.0.15 192.169.0.0 0.0.0.15
Router(config-ext-nacl)# deny ip 192.169.0.16 0.0.0.15 host 192.169.0.35
Router(config-ext-nacl)# permit ip 192.169.0.16 0.0.0.15 any
Router(config-ext-nacl)# exit
```

Apply ACL to HR sub-interface:

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# ip access-group ACL_HR in
Router(config-subif)# exit
```

To Verify ACL:

```
Router# show access-lists
Router# show running-config interface gigabitEthernet 0/0.10
```

```

Router1#show access-list
Extended IP access list VLAN10_Filter
 10 deny ip 192.169.0.16 0.0.0.15 192.169.0.0 0.0.0.15
 20 deny ip 192.169.0.16 0.0.0.15 host 192.169.0.35
 30 permit ip any any (4 match(es))

Router1#

```

Figure X.1 Head Office ACL Configuration

10.2. Branch Office ACL

Branch Office ACLs were configured for GTR-B, GTR-A, Library, Staff, and CCTV VLANs. GTR-B, GTR-A, and Library can communicate with each other but are blocked from Staff, CCTV, Head Office IT, and the File Server.

ACL for GTR-B:

```

Router(config)# ip access-list extended ACL_GTRB
Router(config-ext-nacl)# deny ip 192.168.0.0 0.0.0.31 192.168.0.64 0.0.0.31
Router(config-ext-nacl)# deny ip 192.168.0.0 0.0.0.31 192.168.0.112 0.0.0.15
Router(config-ext-nacl)# deny ip 192.168.0.0 0.0.0.31 192.169.0.0 0.0.0.15
Router(config-ext-nacl)# deny ip 192.168.0.0 0.0.0.31 host 192.169.0.35
Router(config-ext-nacl)# permit ip 192.168.0.0 0.0.0.31 any
Router(config-ext-nacl)# exit

```

Apply ACL to VLAN 30 sub-interface:

```

Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# ip access-group ACL_GTRB in
Router(config-subif)# exit

```

ACL for Staff:

```

Router(config)# ip access-list extended ACL_STAFF
Router(config-ext-nacl)# deny ip 192.168.0.64 0.0.0.31 192.169.0.0 0.0.0.15
Router(config-ext-nacl)# deny ip 192.168.0.64 0.0.0.31 host 192.169.0.35
Router(config-ext-nacl)# permit ip 192.168.0.64 0.0.0.31 any
Router(config-ext-nacl)# exit

```

ACL for CCTV:

```

Router(config)# ip access-list extended ACL_CCTV
Router(config-ext-nacl)# deny ip 192.168.0.112 0.0.0.15 192.169.0.0 0.0.0.15
Router(config-ext-nacl)# deny ip 192.168.0.112 0.0.0.15 host 192.169.0.35
Router(config-ext-nacl)# permit ip 192.168.0.112 0.0.0.15 any
Router(config-ext-nacl)# exit

```

ACLs ensure that unauthorized VLANs cannot access sensitive networks and servers. The report plan also highlights that screenshots of **show access-lists** and the interface where ACL is applied are important proof that ACL is active.

```
Router3#show access-list
Extended IP access list VLAN30_FILTER
 10 deny ip 192.168.0.0 0.0.0.31 192.168.0.64 0.0.0.31 (4 match(es))
 20 deny ip 192.168.0.0 0.0.0.31 192.168.0.112 0.0.0.15 (4 match(es))
 30 deny ip 192.168.0.0 0.0.0.31 192.169.0.16 0.0.0.15 (8 match(es))
 40 deny ip 192.168.0.0 0.0.0.31 192.169.0.0 0.0.0.15 (4 match(es))
 50 deny ip 192.168.0.0 0.0.0.31 host 192.169.0.35 (4 match(es))
 60 permit ip any any (16 match(es))
Extended IP access list VLAN40_FILTER
 10 deny ip 192.168.0.32 0.0.0.31 192.168.0.64 0.0.0.31
 20 deny ip 192.168.0.32 0.0.0.31 192.168.0.112 0.0.0.15
 30 deny ip 192.168.0.32 0.0.0.31 192.169.0.16 0.0.0.15
 40 deny ip 192.168.0.32 0.0.0.31 192.169.0.0 0.0.0.15 (3 match(es))
 50 deny ip 192.168.0.32 0.0.0.31 host 192.169.0.35
 60 permit ip any any (4 match(es))
Extended IP access list VLAN50_FILTER
 10 deny ip 192.168.0.96 0.0.0.15 192.168.0.64 0.0.0.31
 20 deny ip 192.168.0.96 0.0.0.15 192.168.0.112 0.0.0.15
 30 deny ip 192.168.0.96 0.0.0.15 192.169.0.16 0.0.0.15
 40 deny ip 192.168.0.96 0.0.0.15 192.169.0.0 0.0.0.15
 50 deny ip 192.168.0.96 0.0.0.15 host 192.169.0.35
 60 permit ip any any (2 match(es))
Extended IP access list VLAN60_FILTER
 10 deny ip 192.168.0.64 0.0.0.31 192.168.0.112 0.0.0.15
 20 deny ip 192.168.0.64 0.0.0.31 192.169.0.0 0.0.0.15 (4 match(es))
 30 deny ip 192.168.0.64 0.0.0.31 host 192.169.0.35
 40 permit ip any any (8 match(es))
Extended IP access list VLAN120_FILTER
 10 deny ip 192.168.0.112 0.0.0.15 192.169.0.0 0.0.0.15
 20 deny ip 192.168.0.112 0.0.0.15 host 192.169.0.35 (4 match(es))
 30 permit ip any any (8 match(es))
```

Figure X.2 Branch Office matches ACL

```
Router1#show access-list
Extended IP access list VLAN10_Filter
 10 deny ip 192.169.0.16 0.0.0.15 192.169.0.0 0.0.0.15 (8 match(es))
 20 deny ip 192.169.0.16 0.0.0.15 host 192.169.0.35 (4 match(es))
 30 permit ip any any (8 match(es))
```

Figure X.3 Head office matches ACL

XI. User Acceptance Test

After all configurations were completed, connectivity tests were performed using ping and web browser access. Successful tests show that allowed traffic can pass through the network. Failed tests show that ACL rules are correctly blocking unauthorized traffic.

11.1. Head Office Network

- VLAN 10 (HR) is not allowed to communicate with VLAN 20 (IT).

```
C:\>ping 192.169.0.1

Pinging 192.169.0.1 with 32 bytes of data:

Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.

Ping statistics for 192.169.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure XI.1 LAN HR can not ping to LAN IT

- VLAN 10 (HR) is not allowed to access the File Server (192.169.0.35).

```
C:\>ping 192.169.0.35

Pinging 192.169.0.35 with 32 bytes of data:

Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.
Reply from 192.169.0.17: Destination host unreachable.

Ping statistics for 192.169.0.35:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure XI.2 LAN HR can not ping to Server LAN

- VLAN 10 (HR) is allowed to access the Web Server (192.169.0.34) and other permitted networks.

```
C:\>ping 192.169.0.34

Pinging 192.169.0.34 with 32 bytes of data:

Request timed out.
Reply from 192.169.0.34: bytes=32 time=11ms TTL=127
Reply from 192.169.0.34: bytes=32 time<1ms TTL=127
Reply from 192.169.0.34: bytes=32 time=1ms TTL=127

Ping statistics for 192.169.0.34:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms
```

Figure XI.3 LAN HR can ping to Server LAN

- VLAN 20 (IT) is allowed to communicate with VLAN 10 (HR).

```

C:\>ping 192.169.0.17

Pinging 192.169.0.17 with 32 bytes of data:

Reply from 192.169.0.17: bytes=32 time<1ms TTL=255
Reply from 192.169.0.17: bytes=32 time<1ms TTL=255
Reply from 192.169.0.17: bytes=32 time<1ms TTL=255
Reply from 192.169.0.17: bytes=32 time<1ms TTL=255

Ping statistics for 192.169.0.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure XI.4 LAN IT can ping to LAN HR

- VLAN 20 (IT) is allowed to access both the Web Server (192.169.0.34) and File Server (192.169.0.35).

```

Pinging 192.169.0.34 with 32 bytes of data:

Reply from 192.169.0.34: bytes=32 time<1ms TTL=127
Reply from 192.169.0.34: bytes=32 time=1ms TTL=127
Reply from 192.169.0.34: bytes=32 time<1ms TTL=127
Reply from 192.169.0.34: bytes=32 time=20ms TTL=127

Ping statistics for 192.169.0.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 192.169.0.35

Pinging 192.169.0.35 with 32 bytes of data:

Request timed out.
Reply from 192.169.0.35: bytes=32 time=6ms TTL=127
Reply from 192.169.0.35: bytes=32 time=1ms TTL=127
Reply from 192.169.0.35: bytes=32 time<1ms TTL=127

Ping statistics for 192.169.0.35:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

```

Figure XI.5 LAN IT can ping to Web server and File server

- VLAN 20 (IT) is allowed to communicate with all networks and devices on the branch offices.

```
C:\>ping 192.168.0.97

Pinging 192.168.0.97 with 32 bytes of data:

Reply from 192.168.0.97: bytes=32 time<1ms TTL=254
Reply from 192.168.0.97: bytes=32 time<1ms TTL=254
Reply from 192.168.0.97: bytes=32 time<1ms TTL=254
Reply from 192.168.0.97: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure XI.6 LAN IT can ping to LAN Library

```
C:\>ping 192.168.0.113

Pinging 192.168.0.113 with 32 bytes of data:

Reply from 192.168.0.113: bytes=32 time<1ms TTL=254
Reply from 192.168.0.113: bytes=32 time<1ms TTL=254
Reply from 192.168.0.113: bytes=32 time<1ms TTL=254
Reply from 192.168.0.113: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure XI.7 LAN IT can ping to LAN CCTV

```
C:\>ping 192.168.0.65

Pinging 192.168.0.65 with 32 bytes of data:

Reply from 192.168.0.65: bytes=32 time<1ms TTL=254
Reply from 192.168.0.65: bytes=32 time<1ms TTL=254
Reply from 192.168.0.65: bytes=32 time<1ms TTL=254
Reply from 192.168.0.65: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure XI.8 LAN IT can ping to LAN Staff

```

C:\>ping 192.168.0.33

Pinging 192.168.0.33 with 32 bytes of data:

Reply from 192.168.0.33: bytes=32 time<1ms TTL=254
Reply from 192.168.0.33: bytes=32 time<1ms TTL=254
Reply from 192.168.0.33: bytes=32 time=12ms TTL=254
Reply from 192.168.0.33: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms

```

Figure XI.9 LAN IT can ping to LAN GTR-A

```

C:\>ping 192.168.0.01

Pinging 192.168.0.01 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=254
Reply from 192.168.0.1: bytes=32 time<1ms TTL=254
Reply from 192.168.0.1: bytes=32 time=1ms TTL=254
Reply from 192.168.0.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

Figure XI.10 LAN IT can ping to LAN GTR-B

11.2. Branch Office Networks

- VLAN 30 (GTR-B), VLAN 40 (GTR-A), and VLAN 50 (Library) are allowed to communicate with each other (during ping testing only VLAN 30 for VLAN 40 and 50 it will have a result as VLAN 30 due to the same ACL rules).

```

C:\>ping 192.168.0.113

Pinging 192.168.0.113 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.168.0.113:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.11 LAN GTR-B can not ping to LAN CCTV

```

C:\>ping 192.168.0.97

Pinging 192.168.0.97 with 32 bytes of data:

Reply from 192.168.0.97: bytes=32 time=25ms TTL=255
Reply from 192.168.0.97: bytes=32 time=10ms TTL=255
Reply from 192.168.0.97: bytes=32 time=17ms TTL=255
Reply from 192.168.0.97: bytes=32 time=45ms TTL=255

Ping statistics for 192.168.0.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 45ms, Average = 24ms

```

Figure XI.12 LAN GTR-B can ping to LAN Library

```

C:\>ping 192.168.0.65

Pinging 192.168.0.65 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.168.0.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.13 LAN GTR-B can not ping to LAN Staff

```

C:\>ping 192.168.0.33

Pinging 192.168.0.33 with 32 bytes of data:

Reply from 192.168.0.33: bytes=32 time=28ms TTL=255
Reply from 192.168.0.33: bytes=32 time=28ms TTL=255
Reply from 192.168.0.33: bytes=32 time=41ms TTL=255
Reply from 192.168.0.33: bytes=32 time=16ms TTL=255

Ping statistics for 192.168.0.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 41ms, Average = 28ms

```

Figure XI.14 LAN GTR-B can ping LAN GTR-A

- VLAN 30 are allowed to access the Web Server (192.169.0.34).

```

C:\>ping 192.169.0.34

Pinging 192.169.0.34 with 32 bytes of data:

Reply from 192.169.0.34: bytes=32 time=48ms TTL=126
Reply from 192.169.0.34: bytes=32 time=24ms TTL=126
Reply from 192.169.0.34: bytes=32 time=13ms TTL=126
Reply from 192.169.0.34: bytes=32 time=4ms TTL=126

Ping statistics for 192.169.0.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 48ms, Average = 22ms

```

- VLAN 30 are not allowed to communicate with Head Office VLAN 10 (HR) and VLAN 20 (IT).

```

C:\>ping 192.169.0.17

Pinging 192.169.0.17 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.169.0.17:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.169.0.01

Pinging 192.169.0.01 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.169.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.15 LAN GTR-B can not ping to LAN HR and IT

- VLAN 30 are not allowed to access the File Server (192.169.0.35).

```

C:\>ping 192.169.0.35

Pinging 192.169.0.35 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.169.0.35:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.16 LAN GTR-B can not ping to File Server

- VLAN 60 (Staff) is not allowed to communicate with Head Office VLAN 20 (IT).

```

C:\>ping 192.169.0.1

Pinging 192.169.0.1 with 32 bytes of data:

Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.

Ping statistics for 192.169.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.17 LAN Staff can not ping LAN IT

```

C:\>ping 192.169.0.17

Pinging 192.169.0.17 with 32 bytes of data:

Reply from 192.169.0.17: bytes=32 time<1ms TTL=254
Reply from 192.169.0.17: bytes=32 time=1ms TTL=254
Reply from 192.169.0.17: bytes=32 time=1ms TTL=254
Reply from 192.169.0.17: bytes=32 time=1ms TTL=254

Ping statistics for 192.169.0.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

Figure XI.18 LAN Staff can ping to LAN HR

- VLAN 120 (CCTV) is allowed to communicate with Head Office VLAN 10 (HR).

```

Pinging 192.169.0.17 with 32 bytes of data:

Reply from 192.169.0.17: bytes=32 time<1ms TTL=254
Reply from 192.169.0.17: bytes=32 time<1ms TTL=254
Reply from 192.169.0.17: bytes=32 time<1ms TTL=254
Reply from 192.169.0.17: bytes=32 time<1ms TTL=254

Ping statistics for 192.169.0.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure XI.19 LAN CCTV can ping to LAN HR

```

C:\>ping 192.169.0.35

Pinging 192.169.0.35 with 32 bytes of data:

Reply from 192.168.0.113: Destination host unreachable.
Reply from 192.168.0.113: Destination host unreachable.
Reply from 192.168.0.113: Destination host unreachable.
Reply from 192.168.0.113: Destination host unreachable.

Ping statistics for 192.169.0.35:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure XI.20 LAN CCTV cannot ping to File Server

XII. Result and Discussion

The project successfully implemented VLAN segmentation, trunking, inter-VLAN routing, OSPF dynamic routing, Spanning Tree Protocol, server configuration, wireless access, and ACL-based network security.

VLANs were used to separate departments into different logical networks. This improved network organization and reduced broadcast traffic. Trunk ports allowed VLAN traffic to pass

between switches and routers. Inter-VLAN routing allowed devices from different VLANs to communicate through their gateway interfaces.

OSPF was successfully configured between routers. After OSPF convergence, routers learned remote networks dynamically and communication between the Head Office and Branch Office was possible. Spanning Tree Protocol was also configured in the Head Office to prevent switching loops and improve network stability.

The web server and file server were configured in the Server VLAN. The web server was accessible by allowed VLANs, while the file server was protected using ACL rules. The test results showed that allowed traffic was successful and blocked traffic failed, confirming that the security policy worked correctly.

XIII. Conclusion

In conclusion, this project successfully designed and implemented an enterprise network using Cisco Packet Tracer. The network included VLAN segmentation, trunk configuration, inter-VLAN routing, OSPF dynamic routing, Spanning Tree Protocol, server configuration, wireless access, and ACL security rules.

The Head Office and Branch Office were connected through routers using OSPF, allowing automatic route exchange between networks. VLANs separated departments such as HR, IT, Server, GTR-B, GTR-A, Library, Staff, and CCTV. STP/RSTP prevented switching loop the Head Office, while ACLs controlled access between VLANs and protected the File Server from unauthorized users.

Overall, the project helped improve practical skills in routing, switching, VLAN design, security policy implementation, and network troubleshooting. The final testing confirmed that the enterprise network worked correctly and followed the required access control rules.